FAITHFULNESS OF BASE CHANGE ON POWER-TORSION MODULES CHARACTERISES FAITHFUL FLATNESS OF THE QUOTIENT MAP

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The author writes 'I-torsion R-modules' in the proof while the statement says 'I-power torsion'; both phrasings are reproduced as they stand. The three references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- Faithful flatness of a ring map and its equivalence with surjectivity on spectra. invoked at proof lines 24244-24245. Gives the equivalence of conditions (1) and (2).
- Identification of modules annihilated by I with modules over R/I, and the resulting description of their base change. used at proof lines 24246-24252. Gives the equivalence of (1) with condition (3).
- Expressibility of a power-torsion module as a colimit of bounded torsion submodules, and commutation of tensor product with colimits. used at proof lines 24257-24260. Extends faithfulness from the bounded to the power-torsion case, giving (3) implies (4).

Source reference. Original lemma 00HO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 650–650. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00HQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 652–652. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00HI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 644–644. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let $\varphi : R \to S$ be a ring map. Let $I \subset R$ be an ideal. The following are equivalent*

(1) *φ is flat and $R/I \to S/IS$ is faithfully flat,*
(2) *φ is flat, and the map $\mathrm{Spec}(S/IS) \to \mathrm{Spec}(R/I)$ is surjective.*
(3) *φ is flat, and the base change functor $M \mapsto M \otimes_R S$ is faithful on modules annihilated by I, and*
(4) *φ is flat, and the base change functor $M \mapsto M \otimes_R S$ is faithful on I-power torsion modules.*

Proof. If $R \to S$ is flat, then $R/I^n \to S/I^nS$ is flat for every n, see Algebra, Lemma 00HI. Hence (1) and (2) are equivalent by Algebra, Lemma 00HQ. The equivalence of (1) with (3) follows by identifying I-torsion R-modules with R/I-modules, using that

$$M \otimes_R S = M \otimes_{R/I} S/IS$$

for R-modules M annihilated by I, and Algebra, Lemma 00HO. The implication (4) $\Rightarrow$ (3) is immediate. Assume (3). We have seen above that $R/I^n \to S/I^nS$ is flat, and by assumption it induces a surjection on spectra, as $\mathrm{Spec}(R/I^n) = \mathrm{Spec}(R/I)$ and similarly for S. Hence the base change functor is faithful on modules annihilated by I^n. Since any I-power torsion module M is the union $M = \bigcup M_n$ where M_n is annihilated by I^n we see that the base change functor is faithful on the category of all I-power torsion modules (as tensor product commutes with colimits). $\square$